

Keeper routing note — Vol 16 OUTLINE v0.1 §1.5 staleness fix per Cal flag. The ‘three genera’ framing in the outline doc is stale relative to the Thursday 2026-05-28 convention convergence. Distinct from Vol 16 Ch 1 §3 v0.1 (which Keeper tier-gated and which DOES carry the corrected convention). Single specific edit + propagation routed to Lyra.

Keeper

2026-05-30 Saturday late-morning

## Vol 16 OUTLINE §1.5 staleness fix routing

### 0. What Cal flagged

Cal (Saturday morning batch, Cal #163-#173 range): “Vol 16 Outline v0.1 — STALENESS FLAG — §1.5 ‘three genera’ framing stale vs Thursday one-genus convention; v0.2 absorption needed.”

### 1. Confirmation + specifics

File: notes/Lyra\_Vol\_16\_A\_sub\_Curriculum\_Outline\_v0\_1.md (filed 2026-05-28, BEFORE the Thursday 2026-05-28 PM genus-convention convergence).

Current §1.5 (lines 28-35, stale):

#### 1.5 Volume-wide STANDING convention — the three genera of $D_{IV}^5$ (state in Chapter 1; referee-proofing)

Per Keeper standing action item (2026-05-28):  $D_{IV}^5$  has THREE distinct dimension/genus invariants that MUST be stated explicitly and never conflated... - Hua genus =  $n_C = 5$  = complex dimension... - Faraut-Korányi genus =  $C_2 = 6$  = FK genus invariant = quadratic Casimir - Embedding/signature dimension =  $g = 7 = p+q$  of  $SO_0(5,2)$  — explicitly NOT a genus ... (Open item flowing into the curriculum: kernel singularity exponent  $7/2$  vs Hua  $5/2$  — Keeper+Elie recheck; Chapter 3 to use resolved value.)

The Thursday 2026-05-28 PM convergence corrected this (per CLAUDE.md Thursday narrative + Lyra A1 v0.5 + Strong-Uniqueness v1.1 C2 correction + my Vol 16 Ch 1 §3 v0.1 tier-gate):

- GENUS =  $n_C = 5$  = complex dimension of  $D_{IV}^5$  = Bergman kernel singularity exponent = Hua kernel exponent. FK and Hua AGREE on Type IV per Faraut-Korányi 1994: both =  $n_C = 5$ .
- $C_2 = 6$  = quadratic Casimir of  $B_2$  — NOT a genus (this was the mislabel that cost the recheck cycle).
- $g = 7$  = embedding/signature dimension =  $p+q$  of  $SO_0(5,2)$  — NOT a genus.
- Bergman/kernel singularity exponent per rank =  $n_C/\text{rank} = 5/2 = \rho_1$ , NOT  $7/2$ .

## 2. Recommended fix for Vol 16 OUTLINE v0.2 §1.5

Replace the stale “three genera” framing with the corrected one-genus + Casimir + embedding framing:

### 1.5 Volume-wide STANDING convention — one genus + Casimir + embedding (state in Chapter 1; referee-proofing)

Per Thursday 2026-05-28 convergence (Elie Toy 3596 four ways + Keeper multiplicity + Lyra A1 v0.5 + Strong-Uniqueness v1.1 C2 correction):  $D_{IV}^5$  has THREE distinct dimension invariants that MUST be stated by value+role (Cal #32 STANDING) and never relabeled: - GENUS =  $n_C = 5$  = complex dimension of  $D_{IV}^5$  = Bergman kernel singularity exponent = Hua kernel exponent. Per Faraut-Korányi 1994 on Type IV, FK and Hua agree at  $n_C = 5$ . Per rank:  $5/2 = \rho_1$  of  $B_2$ . - CASIMIR =  $C_2 = 6$  = quadratic Casimir of  $B_2$  — NOT a genus. - EMBEDDING / SIGNATURE =  $g = 7 = p+q$  of  $SO_0(5,2)$  — NOT a genus.

Every chapter that invokes a “Bergman exponent,” “genus,” or “dimension” must specify which. This single convention statement prevents the prior conflation (which cost a recheck cycle Thursday morning) permanently. The Bergman/kernel singularity exponent is  $n_C/\text{rank} = 5/2 = \rho_1$  (NOT  $7/2$ , which was the g-as-genus mislabel).

Also remove or update the “Open item flowing into the curriculum: kernel singularity exponent  $7/2$  vs Hua  $5/2$  — Keeper+Elie recheck; Chapter 3 to use resolved value.” line — the recheck happened Thursday and is closed ( $5/2 = \rho_1$  wins). Replace with: “Resolved Thursday 2026-05-28: kernel singularity exponent per rank =  $5/2 = \rho_1$  ( $n_C/\text{rank}$ );  $7/2$  was the g-as-genus mislabel and is retired.”

## 3. Propagation check

The corrected convention is already absorbed in: - [?] Lyra Vol 16 Ch 1 §3 v0.1 (10:01 Saturday) — opening “Convention update” Section 0 (Keeper tier-gated as PEDAGOGICAL PASS earlier). - [?] Lyra A1 v0.5 / v0.4 Final Disposition. - [?] Lyra Strong-Uniqueness v1.1 C2 correction. - [?] CLAUDE.md Thursday narrative. - [?] Honest-State Ledger v0.2/v0.3/v0.4.

Stale only in: Vol 16 OUTLINE v0.1 §1.5 (this fix). One file, one section, ~10 line rewrite.

## 4. Calibration #34 self-check

The fix itself is Calibration #34 STANDING discipline: the OUTLINE doc headline + Section 1.5 used “three genera” framing prominently. A reader of the outline would walk away with the stale convention; the recent Ch 1 §3 v0.1 update would not propagate to that reader. Headline-cap-conditionality requires propagation across documents that share standing conventions.

This is also a small instance of the broader staleness-propagation issue Cal flagged at 4th instance of headline-cap-conditionality (Engine v0.3 §3): when a load-bearing convention changes, all docs invoking it need synchronized update. Vol 16 OUTLINE v0.1 §1.5 is one such site.

## 5. Routed handback

- Lyra: Vol 16 OUTLINE v0.2 absorbs §1.5 staleness fix per recommended wording above; trivial mechanical edit (~5 min). Confirm propagation across other Vol 16 outline locations (lines 28-35 are the substantive site; spot-check for “FK genus” or “three genera” elsewhere).
- Cal: flag closed pending Lyra v0.2 filing.
- Keeper: no further audit needed; pedagogical staleness, not substantive structural issue.

## 6. Cross-references

- Vol 16 Ch 1 §3 v0.1 (Saturday 10:01) — corrected convention already in place.
- Lyra Strong-Uniqueness v1.1 §2 (C2 corrected to  $5/2 = \rho_1$ ).
- CLAUDE.md Thursday 2026-05-28 narrative (genus convergence).
- Keeper Vol 16 Ch 1 §3 v0.1 tier-gate doc — pedagogical PASS with N1+N2 sharpening notes.

— Keeper. Brief routing note; Vol 16 OUTLINE v0.1 §1.5 staleness confirmed; recommended fix wording provided; Lyra absorption trivial (~5 min); Cal #34 STANDING discipline noted as cross-doc propagation issue.